

Speech and Audio Processing

02 - speech production

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慶應義塾大学 高道研究室

Class schedule (tentative)

01	Oct. 03	Introduction	Learn from basic to state-of-the-art about speech processing.
02	Oct. 10	Speech production / 音声生成	
03	Oct. 17	Speech recognition / 音声認識	
04	Oct. 24	Speech synthesis / 音声合成	
05	Oct. 31	Self-supervised learning / 自己教師あり学習	
06	Nov. 7	Foundation model / 基盤モデル	
07	Nov. 14	Sound recognition/synthesis / 環境音認識合成	
08	Nov. 28	Advanced topics / 発展的課題	
09	Dec. 05	Advanced topics / 発展的課題	
10	Dec. 12	Speech system building / 音システム構築	Learn how to build your speech system.
11	Dec. 19	Speech system building / 音システム構築	
12	Dec. 26	Speech system building / 音システム構築	
13	Jan. 09	Speech system building / 音システム構築	
14	Jan. 23	Final presentation / 最終発表	

About this class

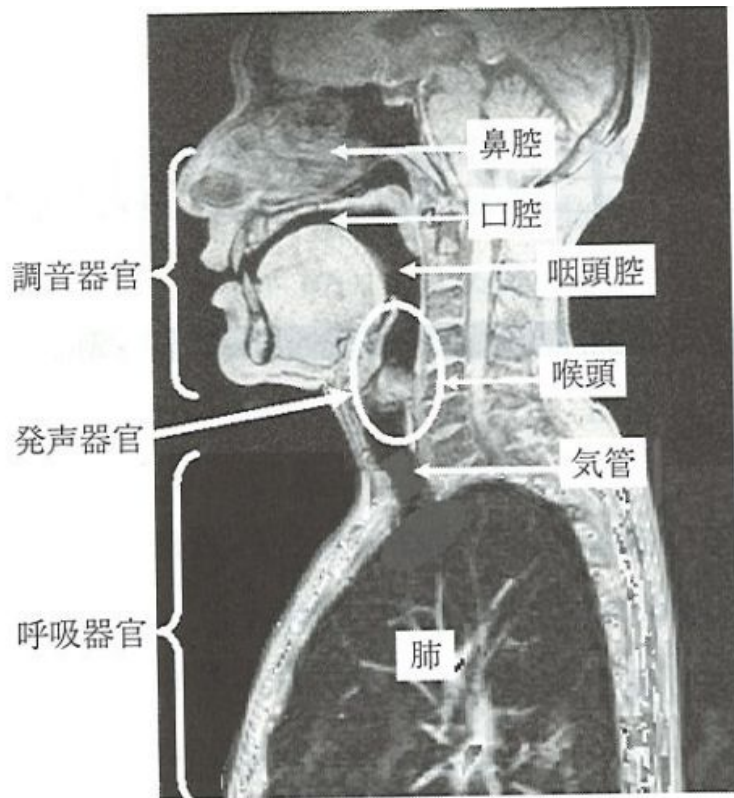
- Language
 - Slides in English (w/ partial translation to Japanese, e.g., named entity)
 - Talk in Japanese (w/ simultaneous translation to English.)
 - Once translation finished for current slide, I will move onto next slide.
 - Either En or Ja is fine for your question.
- Grades
 - 50 / 100 ... final presentation
 - 50 / 100 ... final report
- Room
 - 14-202

Final presentation & reports / 最終発表&レポート

- Perform a unique project that either
 - improves on state-of-the-art in speech-related {signal processing, machine learning} tasks,
 - applies {machine learning, signal processing} to a unique task,
 - investigates human behaviors in human-speech-AI interaction.
 - prototypes the future of human and speech AI.
- Presentation & report
 - Slides in English.
 - Talk in Japanese or English. (choose one you prefer.)
 - I will drive {Ja, En}-to-{En, Ja} simultaneous translation during presentation.
 - Reports in Japanese or English. (choose one you prefer.)

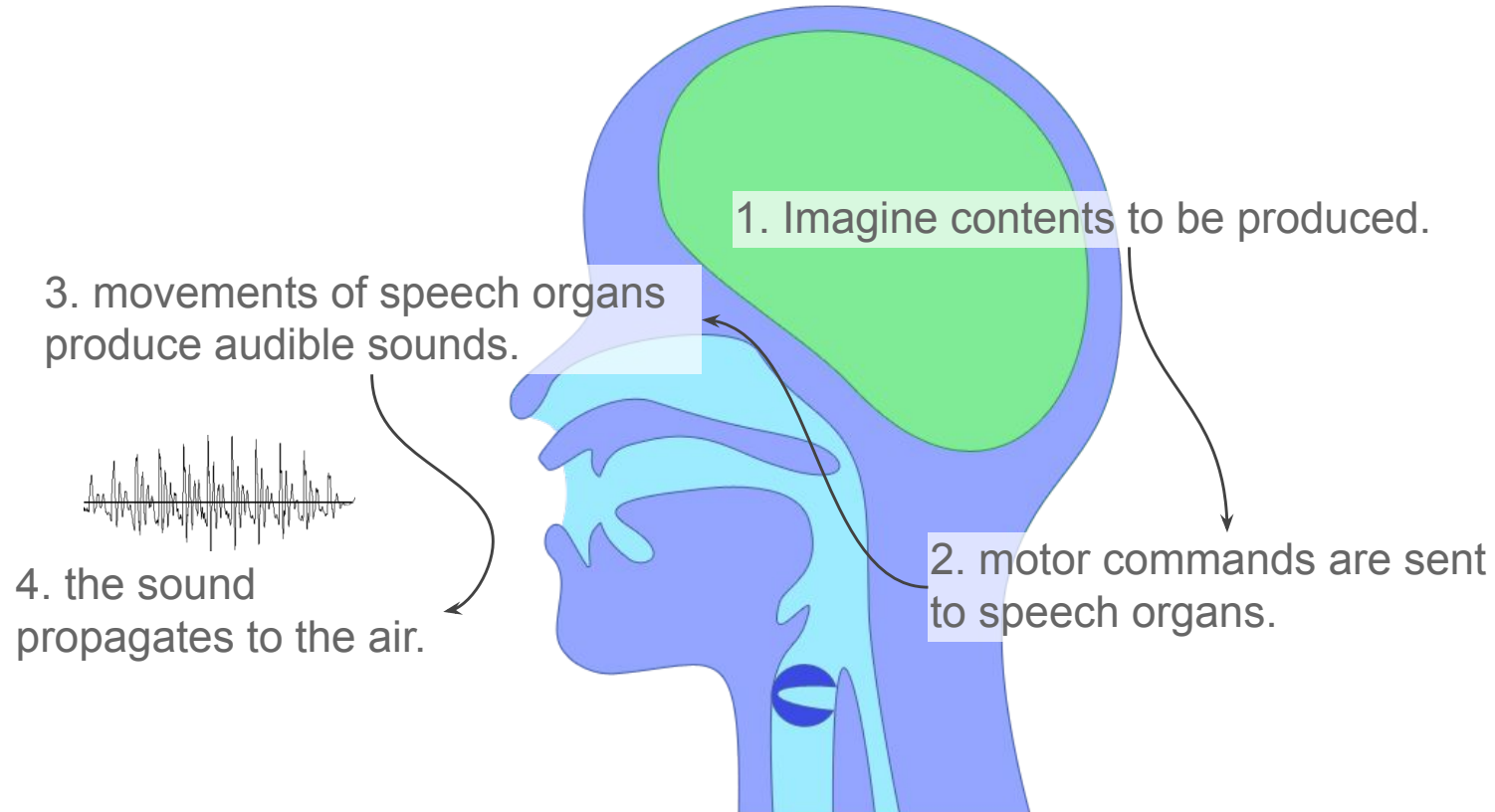
Today's goal

What happens in our body when producing speech

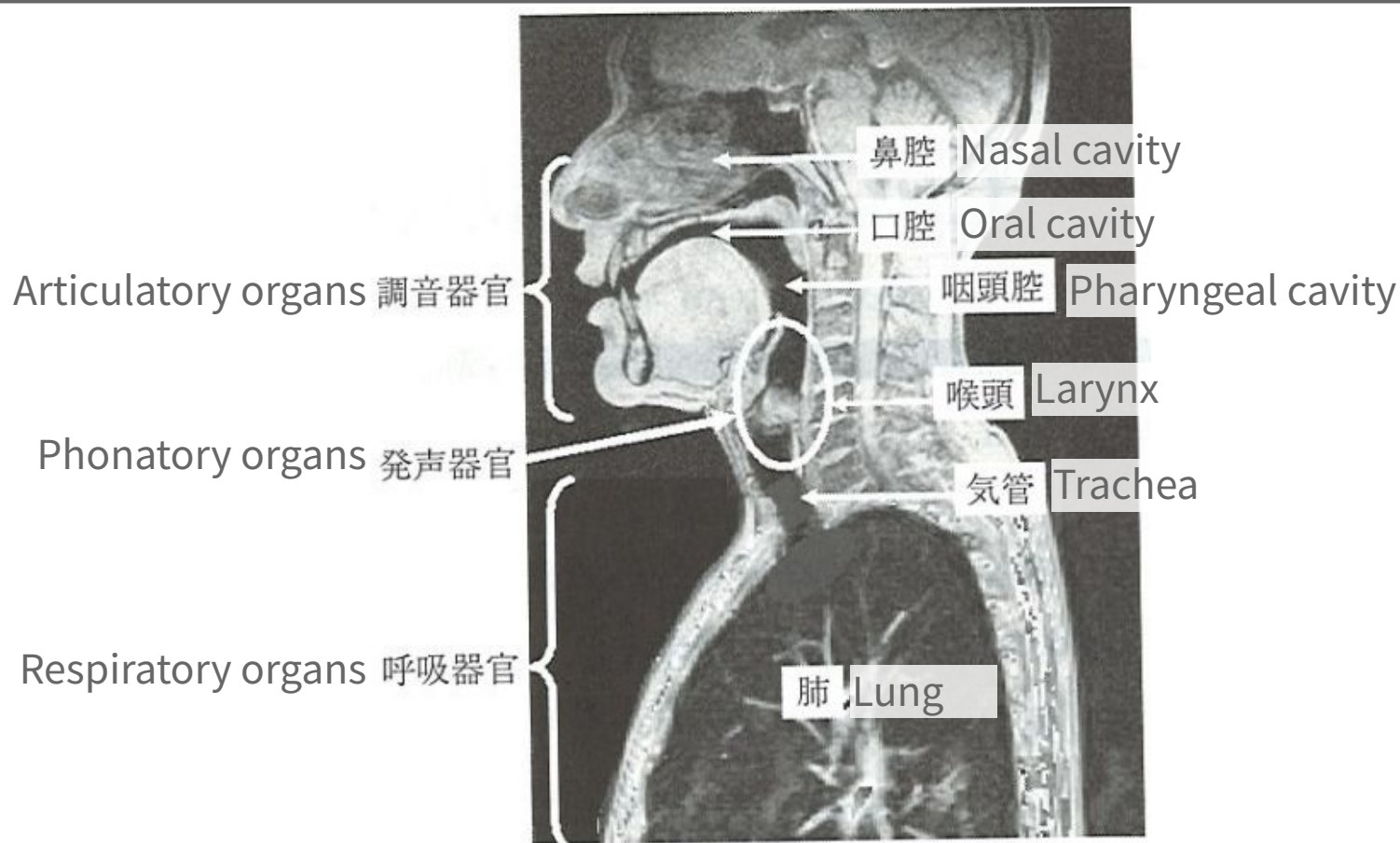


Speech production / 音声生成

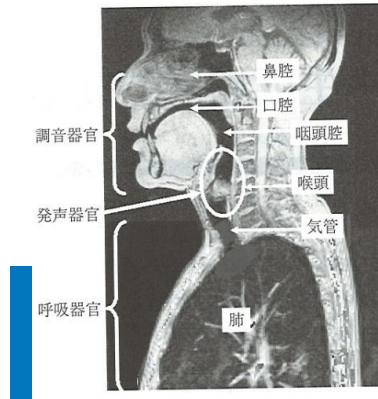
Speech = human voice



Speech organs / 音声器官



Respiratory organs / 呼吸器官



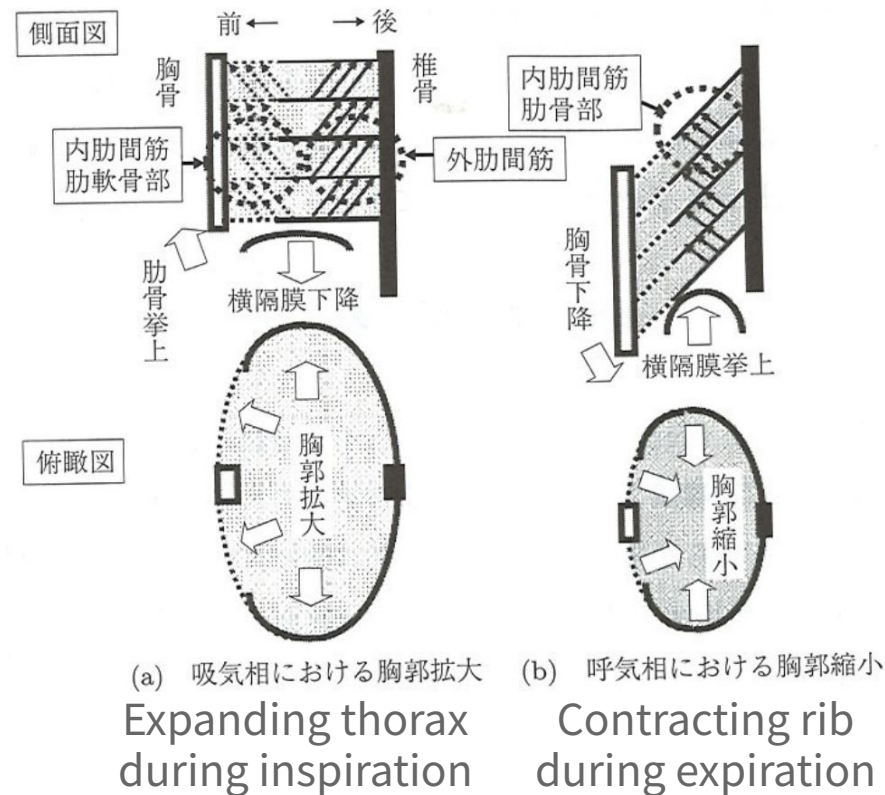
Respiratory organs / 呼吸器官

- **Inspiration (吸気)**

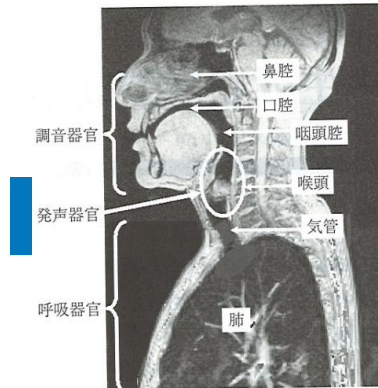
- Sending air to lungs
- The diaphragm descends and the thorax expands. / 横隔膜の下降と胸郭の拡大

- **Expiration (呼気)**

- Expelling air from the lungs
- The diaphragm rises and the internal intercostal muscles of the rib portion contract. / 横隔膜の挙上と内肋間筋の肋骨部の収縮



Phonatory organs / 発声器官



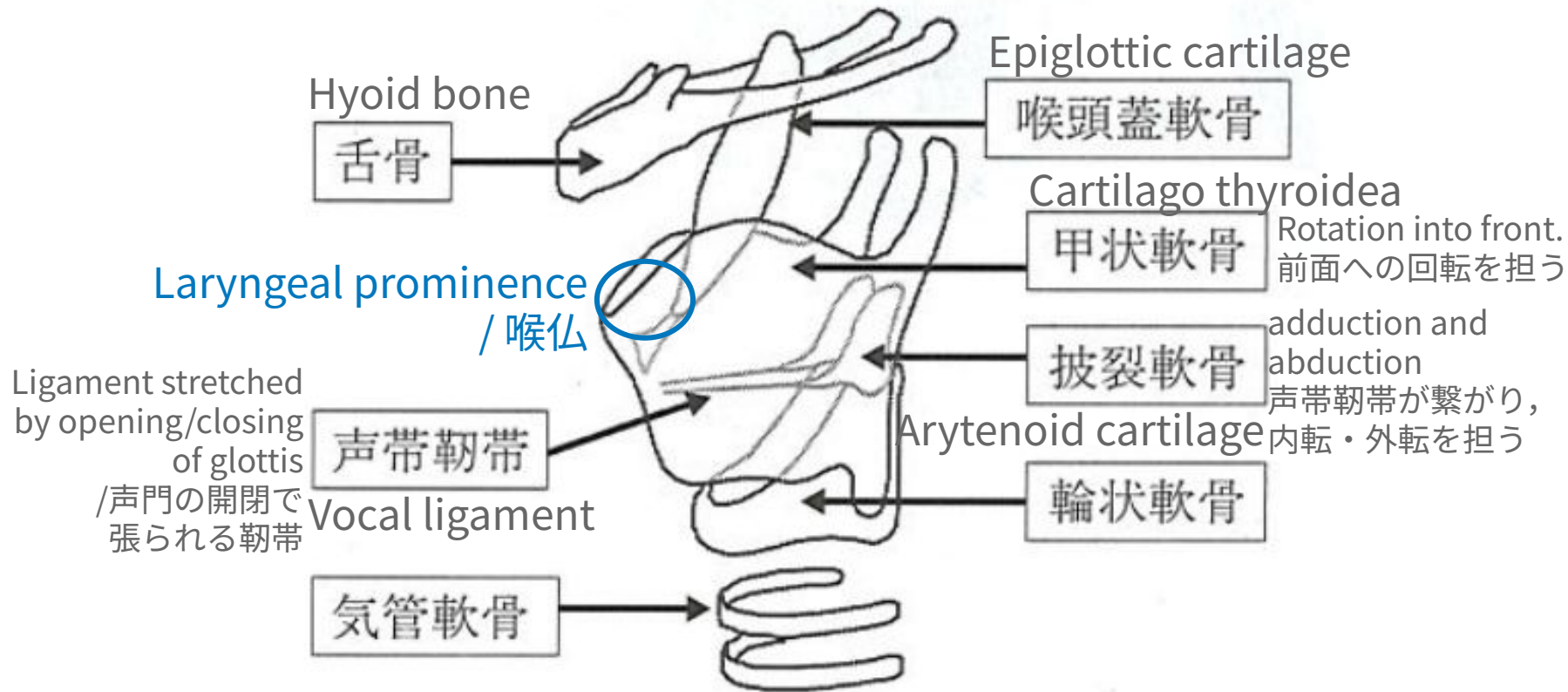
Phonatory organs / 発声器官



Making voicing/unvoicing (w/ or w/o vibration) sounds by this air valve. 12

Phonatory organs / 発声器官

Flap that guides food into the esophagus
食べ物を食道に導く蓋



Physical model around larynx / 喉頭近辺の物理モデル



View from front

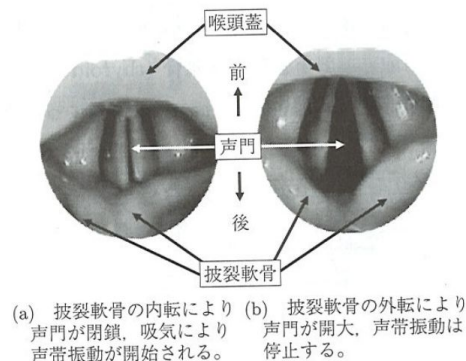
View from back



Generation of burst and turbulent noise by abduction of the arytenoid cartilages / 披裂軟骨の外転による破裂音源・乱流雑音源の生成

• Arytenoid cartilage / 披裂軟骨

- Glottis is closed by adduction / 内転で声門が閉鎖
- Glottis is open by abduction / 外転で声門が開大



• Semi-periodic open-close movements / 準周期的な開閉運動

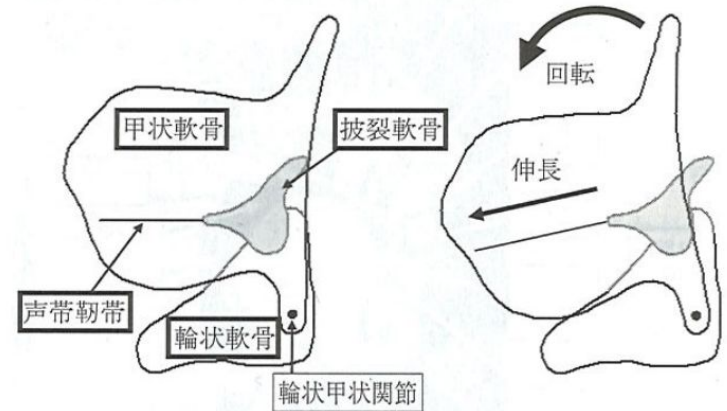
- Adduction closes glottis / 内転により声門が閉鎖
- Air pressure in subglottis becomes greater than supraglottis / 肺からの呼気により声門下の圧力が, 声門上より大きくなる
- The pressure difference causes the glottis to open, allowing air to flow above the glottis / 圧力差により声門が開き, 少量の空気が声門上へ

Fundamental frequency / 基本周波数

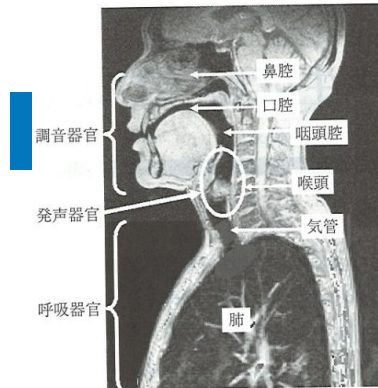
- Inverse of the period of open-close movements / 開閉運動の周期の逆数
 - Roughly equal to “pitch” / いわゆる”声の高さ”
 - Around 100 Hz for adult males, 200 Hz for adult females / 男性の場合は 100 Hz前後, 成人女性の場合は 200 Hz前後
 - The symbol F_0 (エフゼロ) is used.
- Note
 - The fundamental frequency does not always exist. / 基本周波数は常に存在するわけではない.
 - In the case of turbulent noise, the fundamental frequency is absent. / 後述する破裂音源・乱流雑音源では, 基本周波数は存在しない.
 - It is treated as 0 Hz in computers but does not mean 0Hz in physics. / コンピュータでは 0 Hz と扱うこともあるが, 物理的に 0 Hz ではない

How we control the fundamental frequency? / 基本周波数をどうやって変えているのか？

- Rotation of cartilago thyroidea (輪状軟骨) and arytenoid cartilage (甲状軟骨)
 - Angle by two becomes larger (moving front side)
 - Vocal ligament stretched tightly, increasing the stiffness of vocal folds. / 声帯靱帯が強く張られ、声帯の剛性が上がる
 - The natural frequency of the vocal folds becomes higher → the fundamental frequency increases. / 声帯の固有周波数が高くなる = 基本周波数が上がる
 - In the opposite process, the fundamental frequency decreases. / 角度が小さくなると基本周波数は下がる



Articulatory organs / 調音器官



Articulatory organs / 調音器官

日本語の「ワ」と「ファ」

「ワ」と「ファ」の発音の仕組みを、リアルタイムMRI映像を用いて解説します。舌の動きや歯茎との接触など、発音の瞬間を捉えることが可能です。



耳内構造と耳道マイクロビュー



東京話の「ワ」「ファ」の子音群





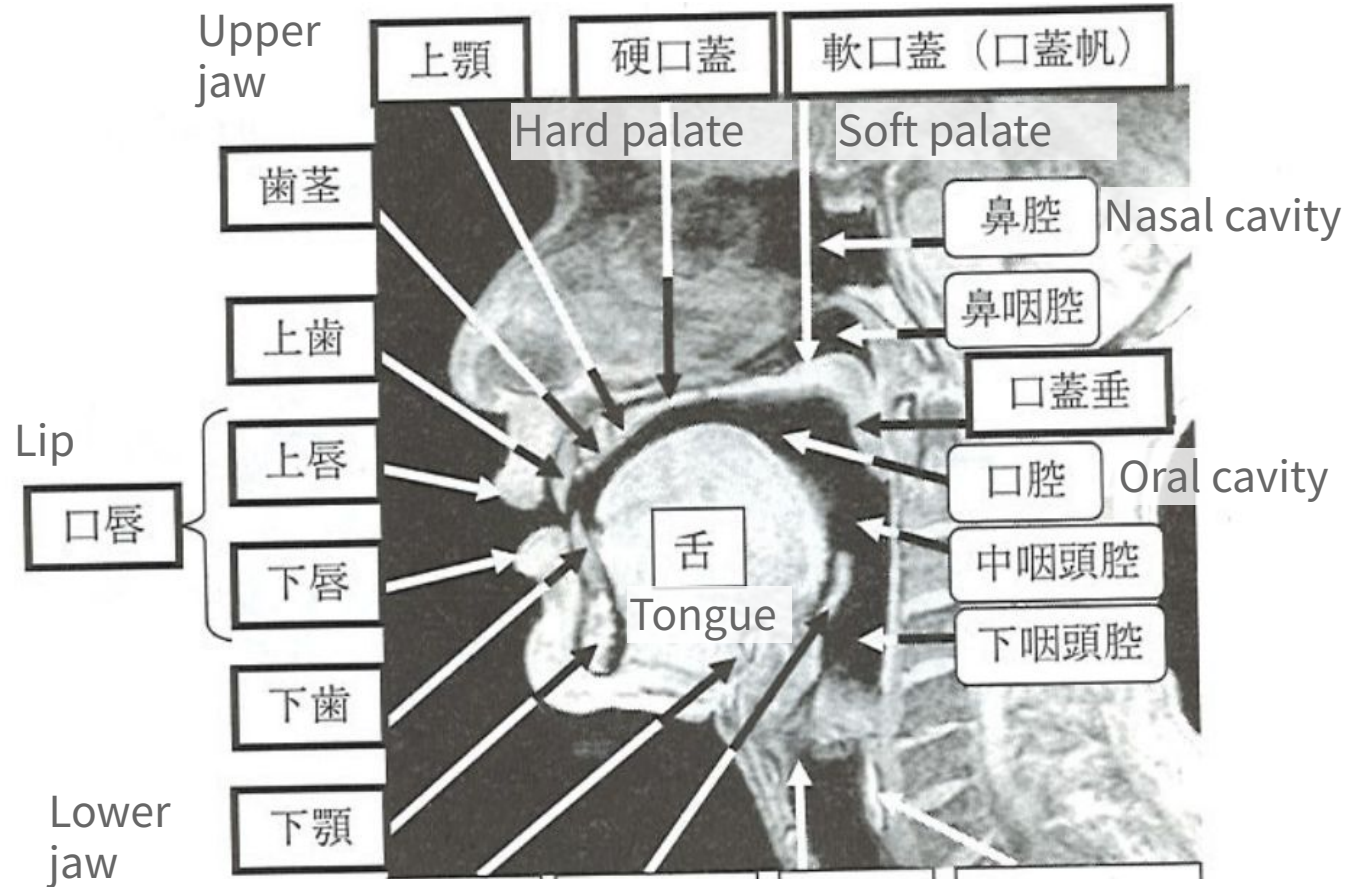
NINJAL
フォーラム

講演

リアルタイムMRI 動画を用いた 調音音声学の再構築 —ワ行子音の問題—

講師：前川喜久雄

Vocal tract adds timbre to sound source. / 声道が音源に音色をつける



Jaws shapes the oral cavity. / 顎が口腔を形作る

- **Upper jaw / 上顎 (じょうがく)**

- Hard palate (硬口蓋): hard parts in palate
- Soft palate (軟口蓋): soft parts in palate
 - It also serves to separate the nasal cavity from the oral cavity.
- alveolar process (歯槽突起)
- Alveolar ridge (歯茎突起, はぐきとつき): upper teeth

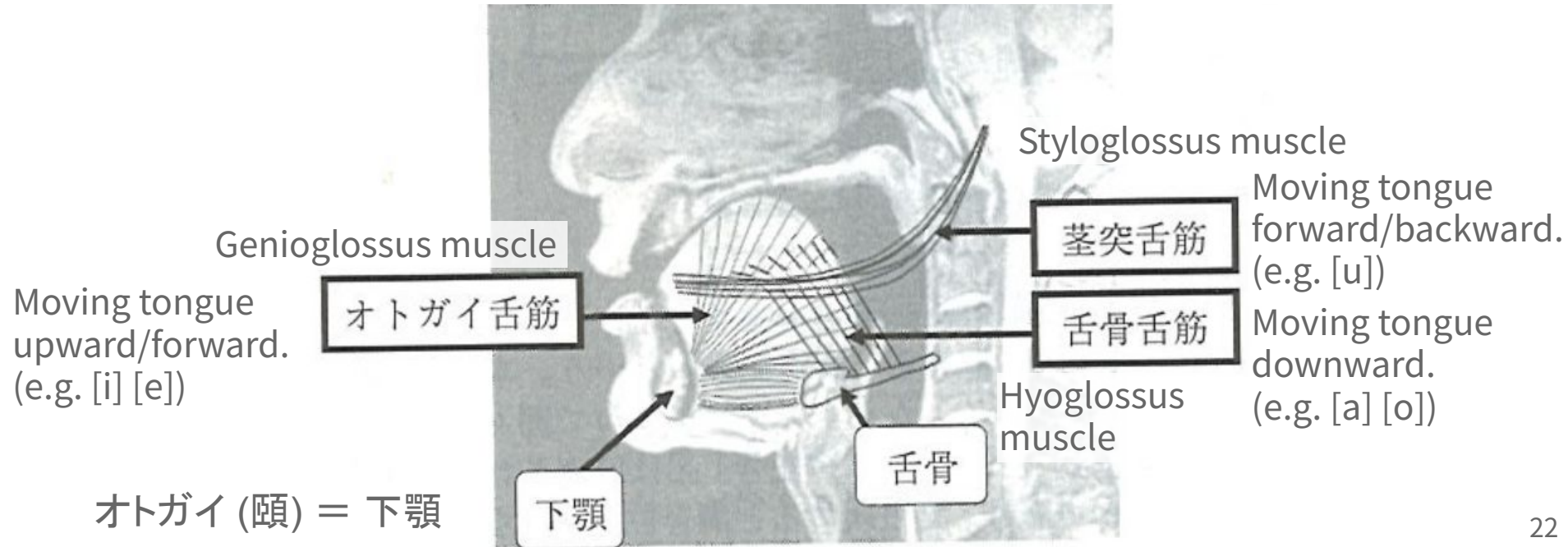
- **Lower jaw / 下顎 (かがく)**

- Rotation around the temporomandibular joint / 顎関節を軸とする回転
- Control of tongue position / 舌位置の制御

Tongue is most important in articulation. / 舌が調音に最も寄与する

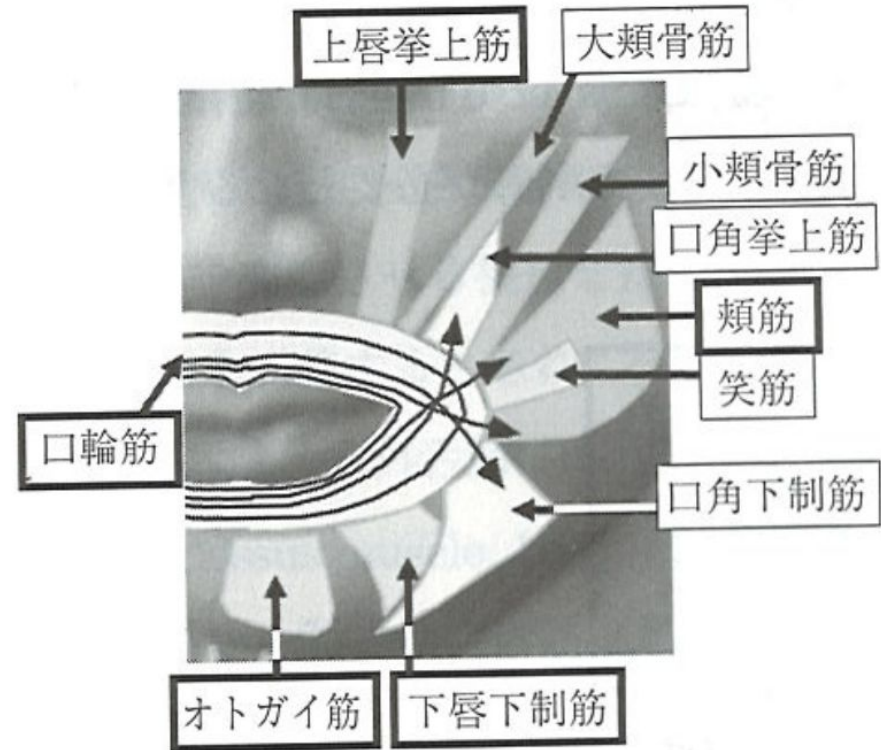
- **Extrinsic tongue muscles / 外舌筋 (がいぜつきん)**

- Originating from the surrounding bones, these muscles deform the entire tongue / 周囲の骨に起始を持ち，舌全体を変形させる



Lips make stop, constriction, and protrusion. / 口唇は閉鎖，狭め，突き出しを行う

- Stop:
 - [p], [b], [m]
- Constriction and protrusion:
 - [ɸ] (“hu”), [w]
- Controlled by orbicularis oris muscle (口輪筋) and mentalis muscle(オトガイ筋)

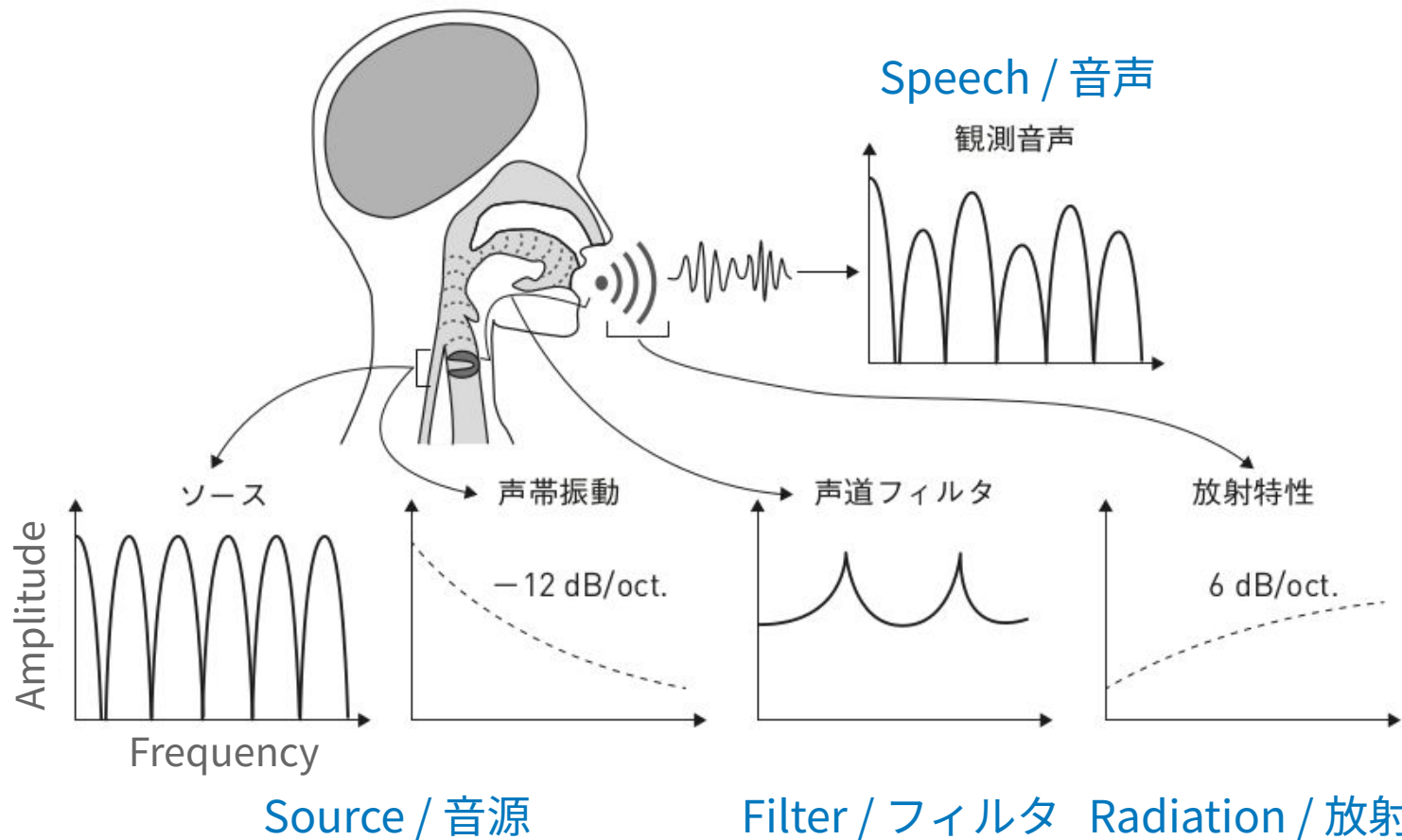


Source-filter theory / ソース・フィルタ理論

History / 歴史

- Developed by Fant
 - Fant, “Acoustic Theory of Speech Production”, 1960.
 - **Cascading three linear models:**
 - Source by vocal cords / 声帯による音源
 - Filter by vocal tract / 声道によるフィルタ
 - Radiation via lip / 口唇からの放射
- Originated from Chiba (千葉) and Kajiyama (梶山)
 - Chiba and Kajiyama, “The Vowel: Its Nature and Structure”, 1941.
 - This paper introduced “production” (生成) and “articulation” (調音).

Source-filter model / ソース・フィルタモデル



Vowels and consonants / 母音と子音

Vowels and consonants in linguistic sounds

/ 言語音における母音・子音

- Phoneme / 音素
 - Smallest units that make one word different from another.
/ 意味の違いをもたらす, 音韻的な最小単位
- Vowel (母音, V)
 - Articulatorily: Sounds without closure involving the tongue, lips, or teeth. / 調音的には「舌, 唇, 歯での閉鎖を伴わない」音
 - Phonologically: The nucleus of a syllable that can form a syllable alone or in combination with consonants. / 音韻的には「単独あるいは子音を伴って音節を構成する」主音
- Consonant (子音, C)
 - Phonemes other than vowels / 母音以外の音

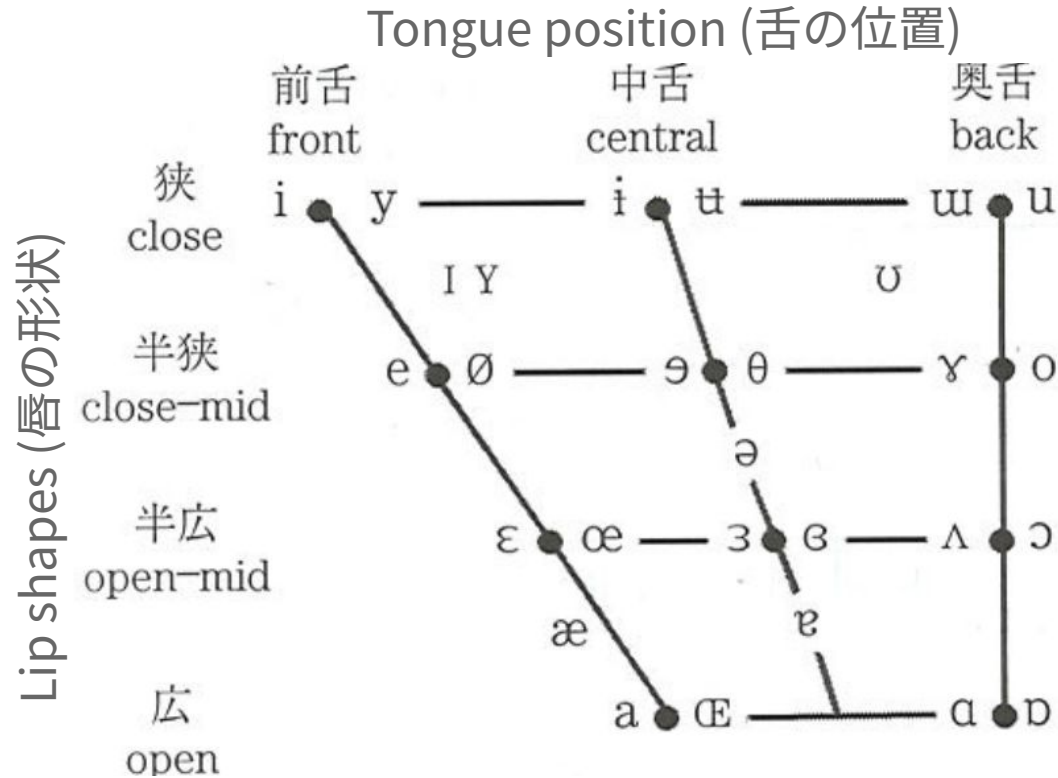
International Phonetic Alphabet, IPA

/ 国際音声記号

- International Phonetic Association (国際音声学会)
 - Academic society related to phonetics / 音声学に関連する学術団体.
- IPA
 - Universal symbols to describe every kinds of languages. / あらゆる言語の音声を統一的に表記するための音声記号
 - <https://www.internationalphoneticassociation.org/content/ipa-chart>

Classification of vowels by IPA

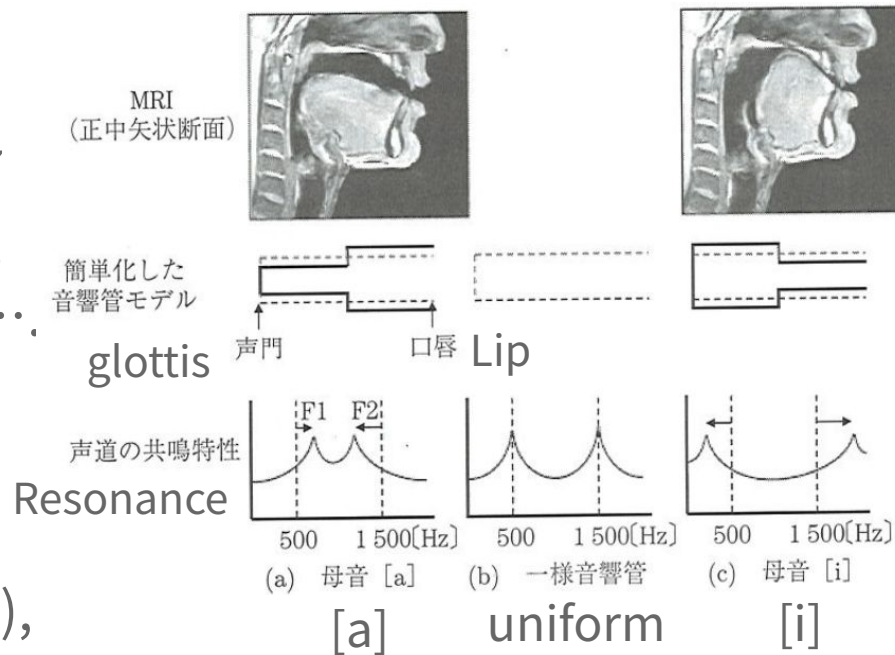
/ IPA による母音の分類



Left is unrounded lip, right is rounded lip
左は非円唇母音, 右は円唇母音である

Formant frequency / フォルマント周波数

- The vocal tract can be seen as concatenation of acoustic tubes.
- If the uniform acoustic tube is 17 cm long (average among males), it resonates at 500 Hz, 1500 Hz, ...
- The frequencies of these resonances are called formant frequencies (フォルマント周波数), denoted as F_1 , F_2 , ... in ascending order.



Classification of consonants by IPA

/ IPA による子音の分類

調音位置 調音様式	両唇音 bilabial	唇歯音 labiodental	歯音 dental	歯茎音 alveolar	後部歯茎音 post alveolar	そり舌音 retroflex	硬口蓋音 palatal	軟口蓋音 velar	口蓋垂音 uvular	喉頭音 pharyngeal	声門音 glottal
破裂音 plosive	p b		t d			ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
鼻音 nasal	m	ɱ	n			ɳ	ɲ	ŋ	ɴ		
ふるえ音 trill	ʙ		r						ʀ		
弾き音 tap or flap		ɸ	ɾ			ɽ					
摩擦音 fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
側面摩擦音 lateral fricative			ɬ ɮ								
接近音 approximant		ʋ	ɹ			ɻ	j	ɰ			
側面接近音 lateral approximant			l			ɭ	ʎ	ʟ			

1つのセルに2つの記号がある場合、左側が無声音、右側が有声音の記号である。
Left is unvoiced consonant, right is voiced consonants.

Conclusion / まとめ

Conclusion

- Speech organs / 音声器官
 - Respiratory organs / 呼吸器官
 - Phonatory organs / 発声器官
 - Articulatory organs / 調音器官
- Source-filter theory / ソース・フィルタ理論
 - Source by vocal cords / 声帯による音源
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